

Deep Trap Origins, Characteristics, and Related Mechanisms in Chemically Grafted Polypropylene with Enhanced Direct Current Volume Resistivity

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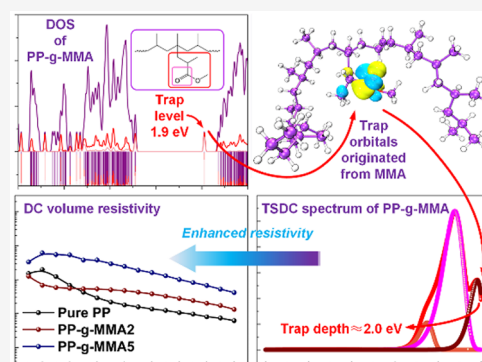
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ABSTRACT: Deep traps play an important role in the high-temperature dielectric properties enhancement of chemically grafting-modified polypropylene (PP), which is fairly promising in the field of insulation material application. This study quantitatively investigates the deep traps of methyl methacrylate (MMA)-grafted PP from three points of view: thermally stimulated depolarization current (TSDC), quantum chemistry computation, and direct current (DC) volume resistivity. The TSDC results indicate that grafted MMA groups can deepen the original trap level of PP and introduce a new deep trap level at about 2.0 eV. The trap origin and positive effect on macroscopic DC resistivity are analyzed and verified by quantum chemistry computation and space charge limited current (SCLC) theory, respectively. We interpret the deep trap of MMA-grafted PP from different perspectives and try to establish the correlations among trap origins, characteristics, and macroscopic manifestation. This work is expected to advance the understanding on trap characteristics of grafting-modified PP and promote the related mechanism research.



1. INTRODUCTION

Regarded as a promising insulation material for the next generation power system, polypropylene (PP) has been extensively studied and investigated in recent years, because of its high thermo-stability, excellent dielectric performances, and thermoplastic nature, which means it is recyclable and eco-friendly.^{1–3} Grafting modification by functional chemical groups has been proved to be a fairly effective method to further enhance the dielectric properties, especially for direct current (DC) insulation under high temperature.^{4,5} It can achieve the same enhancement level as for the PP-based nanocomposite, but it is free of the agglomeration problem as exists in the latter.^{6,7} Therefore, grafting-modified PP is very suitable for large-scale industrial production for the insulation of practical high-voltage apparatuses, such as high voltage direct current (HVDC) extruded cables⁷ and metallized film capacitors.⁸

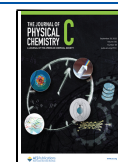
A primary issue on PP-based grafting modification is how does the grafted chemical group enhance the dielectric properties of PP, that is, the mechanism, which is vital, and rather interesting. Certainly, some pioneering investigations have been carried out on this issue, and corresponding rudimentary theories have emerged. Dang et al.⁶ and He et al.⁹ attempted to grafting-modify PP by maleic anhydride (MAH), which is a polar group with two carbonyls in the chemical structure, and realized a significant enhancement on DC

dielectric properties, including enhanced DC volume resistivity and DC breakdown strength, suppressed space charge accumulation, which is very beneficial to the application in DC cables. Related experimental characterizations indicate that grafting modification can affect the crystallization morphology of PP,^{4–6} thus further affecting the macroscopic properties. Furthermore, what is more arrestive, the thermally stimulated depolarization current (TSDC) results implicate large quantities of deep traps are introduced in the PP matrix after grafting modification.^{9–11} Considering this phenomenon is similar to that in PP-based nanocomposites,^{12,13} one can speculate the trap theory of nanodielectrics may also be available in grafting-modified PP, and the key point might be the grafted polar group. However, the in-depth relationship between the chemical structure of the polar group and the trap characteristics, as well as the connection between grafting-introduced traps and the dielectric performance of grafted PP, are not clear yet, nor investigated.

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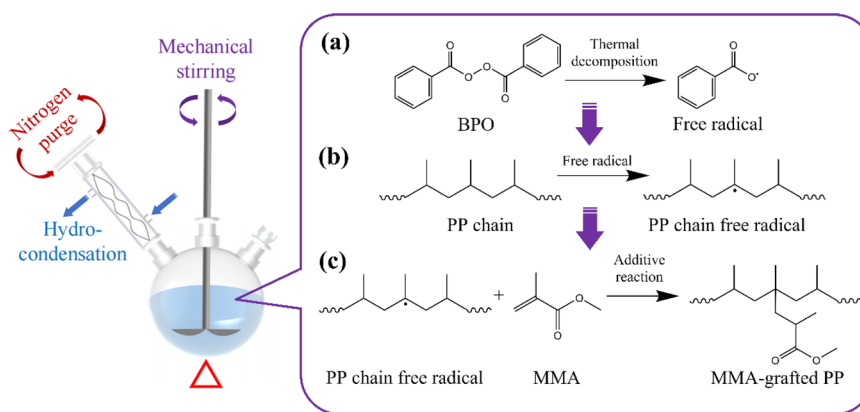


Figure 1. Schematic diagram of grafting reaction: (a) initiator BPO thermally decomposes into free radical; (b) PP chain free radical is generated after hydrogen seized; (c) MMA is additively grafted onto PP chain.

In this paper, we select a typical polar grafting monomer methyl methacrylate (MMA) to grafting-modify PP. The deep trap characteristics are quantitatively investigated by TSDC experiment, quantum chemistry computation, and DC volume resistivity based on space charge limited current (SCLC) theory. The main purpose of this study is to develop and advance the understanding of the deep trap characteristics in grafting-modified PP, trying to establish the relationships among the chemical structure of grafted group, trap characteristics, and macroscopic dielectric properties. This would also be beneficial to the mechanism research of grafting modification and following industrial production.

2. EXPERIMENTAL SECTION

2.1. Materials and Preparation. Isotactic polypropylene with the density of 0.9 g/cm³ is chosen as the PP matrix, provided by Maoming Petrochemical. Grafting monomer MMA is purchased from Beijing J&K Scientific Ltd. and purified by vacuum distillation before reaction. Benzoyl peroxide (BPO, Beijing J&K Scientific Ltd.) is adopted to initiate the grafting reaction. Figure 1 exhibits the grafting reaction mechanism: BPO thermally decomposes into free radical (black dot), which can seize the hydrogen atom of PP molecular chain, thus generating the PP chain free radical. MMA can additively react at the free radical site thereby grafted onto the PP chain.

The reaction procedure: the PP matrix is first vacuum-dried under 70 °C for 24 h and then put into a reactor. Pure nitrogen is purged into the reactor to remove the air. The weighed BPO and MMA are mixed into a uniform solution and then added into the reactor. The MMA weight percent (to the weight of PP matrix) is controlled as 2% and 5%, respectively. The reaction starts after the temperature stabilized at 90 °C, with even mechanical stirring for 5 h. After reaction, the reactor is repurged by nitrogen. The reacted mixture is taken out and then filtered and dried. The obtained grafting-modified PPs are respectively named PP-g-MMA2 and PP-g-MMA5. Furthermore, to remove the self-polymerized MMA that may be generated during reaction, the Soxhlet extraction is employed to purify the prepared PP-g-MMA by ethyl acetate for 24 h. The purified materials would be free of the influence of self-polymerized MMA. The mass loss of MMA after extraction is approximately 7%, indicating that the grafting modification is successful and sufficient.

For following experimental characterizations, the modified PPs are prepared into film samples: the weighed material is melted for 5 min at 200 °C, then a 15 MPa pressure is applied for 10 min, and finally the material is hydro-cooled to room temperature under 10 MPa for 5 min. Unmodified PP film sample is also prepared as the reference, denoted as pure PP.

Fourier transform infrared spectroscopy (FT-IR, Thermo Fisher Nicolet iS10, USA) is utilized to characterize the grafted MMA group. The scanning mode is transmission, with the scanning resolution of 4 cm⁻¹, and scanning range 400–4000 cm⁻¹. The FT-IR spectra of the three samples are normalized and shown in Figure 2. An obvious absorption peak at 1729

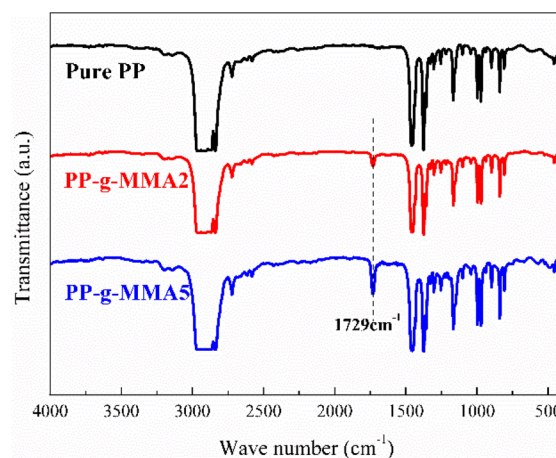


Figure 2. FT-IR spectra of MMA-grafted PP and pure PP.

cm⁻¹ can be observed with proportional amplitude to the grafting content, which corresponds to the stretch vibration of the carbonyl group (C=O) of grafted MMA. Therefore, it can be concluded that the MMA group is successfully grafted onto the PP molecular chain.

2.2. Experimental Design. The TSDC test is carried out by a broadband dielectric spectrometer (NovoControl GmbH, Germany). The sample is of 50 μm average thickness and sputter-coated with gold electrodes before testing. The schematic diagrams of TSDC circuit and test procedure are presented as Figure 3a,b. At first the sample is polarized under a 20 kV/mm DC electric field for 30 min at 110 °C. Then it is rapidly cooled to −100 °C by liquid nitrogen to freeze the trapped charges. Afterward the sample is short-circuited for 15

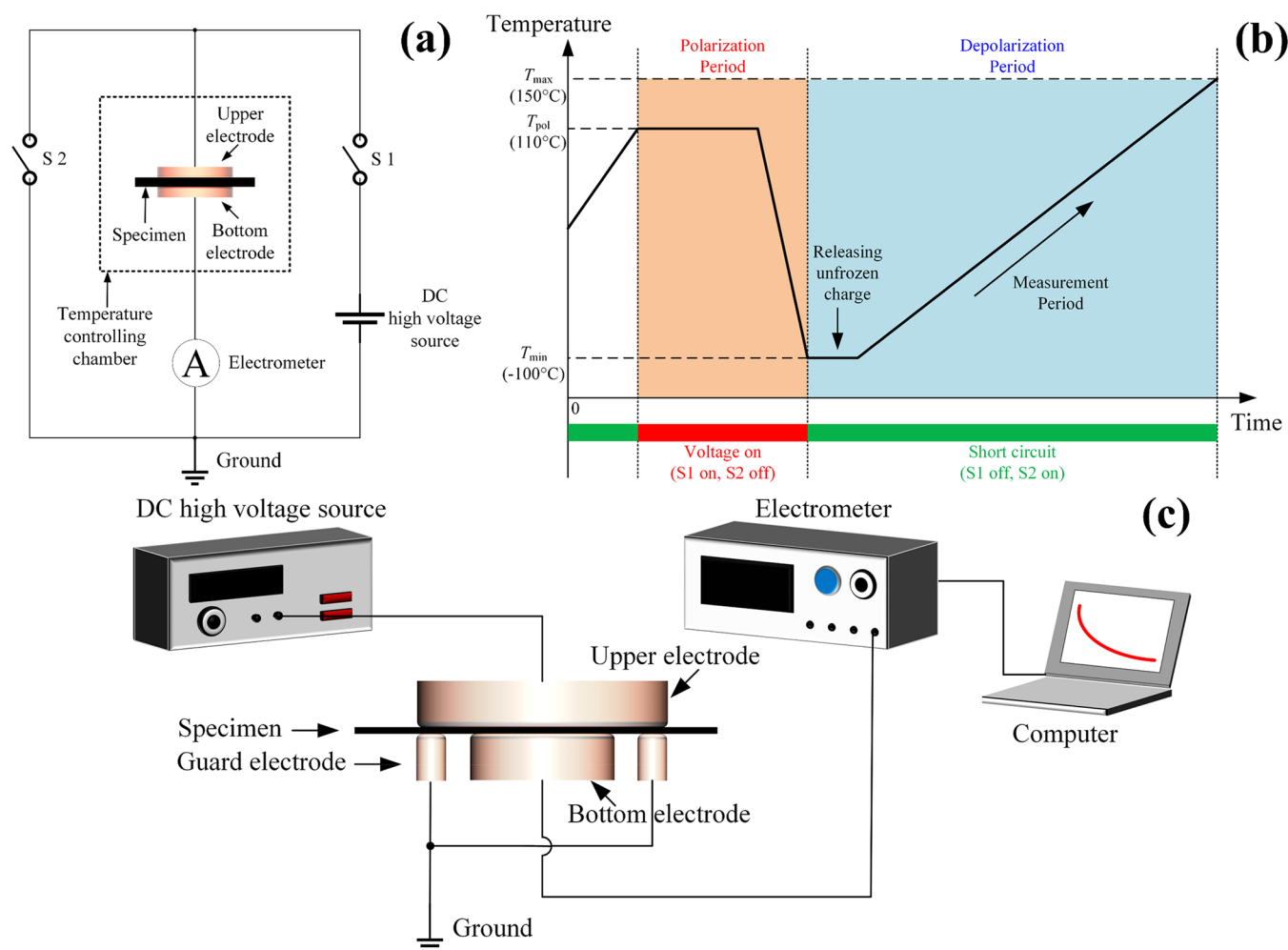


Figure 3. Experimental design: (a) TSDC circuit; (b) TSDC test procedure; (c) test system of DC volume resistivity.

min to release the unfrozen charges and then reheated to 150 °C at 3 °C/min to measure the depolarization current by an electrometer (Keithley 6517B, USA). On the basis of the single trap current equation of TSDC under high electric field (1)¹⁴

$$I(t) = A \exp \left[-\frac{E_t}{kT} - \frac{\nu}{\beta} \int_{T_0}^T \exp \left(-\frac{E_t}{k\tau} \right) d\tau \right] \quad (1)$$

where t is time (s), A is the pre-exponential factor, E_t is the trap depth (eV), k is the Boltzmann constant (J/K), ν is the frequency factor (s^{-1}), T is the absolute temperature (K), T_0 means the initial depolarization temperature, and β is the heating rate (K/s) ($T = T_0 + \beta t$), the trap peaks in TSDC spectra are algorithmically separated and the corresponding trap parameters are extracted to realize the quantitative trap analysis.¹⁵

DC volume resistivity is measured by a three-electrode platform at 70/90/110 °C. Figure 3c shows the test system. The samples are of around 100 μm thickness, also sputter-coated as in TSDC test. The applied electric field rises from 5 to 80 kV/mm by 5 kV/mm steps. In each period, the stabilized DC leakage current is recorded and utilized to calculate the DC volume resistivity according to IEC Standard 60093. Moreover, the leakage current density–electric field (J – E) characteristic is plotted in the log–log coordinate and analyzed on the basis of SCLC theory.^{16,17}

2.3. Computational Details. A PP segment consisting of 12 propylene monomers is established in Gaussian 16W software. One MMA group is grafted onto the segment to simulate the MMA-grafted PP segment. The segment conformation is optimized and the electron wave functions are calculated on the basis of the B3LYP functional and 6-31G* basis function.^{10,18} On the basis of the wave function calculation results, the density of electron states (DOS), lowest unoccupied molecular orbitals (LUMO), and surface electrostatic potential (ESP) of the segment are quantitatively calculated.¹⁹ Besides, the noncovalent interactions between the grafted MMA group and the PP chain are analyzed by reduced density gradient (RDG) theory.²⁰

3. RESULTS AND DISCUSSION

3.1. TSDC Analysis. The original TSDC spectra are shown in Figure 4a. One can easily observe the complicated peak distribution in grafting-modified PP. The peak amplitudes in the high-temperature range (over 50 °C) significantly differ from those of low temperature (below 0 °C), which should be ascribed to the different origins. As shown in the inner semilogarithmic figure of Figure 4a, the peak at about –10 °C corresponds to the glass transition of PP,⁷ whereas the nonzero depolarization current in the range –100 to –50 °C should be attributed to the relaxation of MMA side group, whose amplitude is proportional to MMA content. This wide current

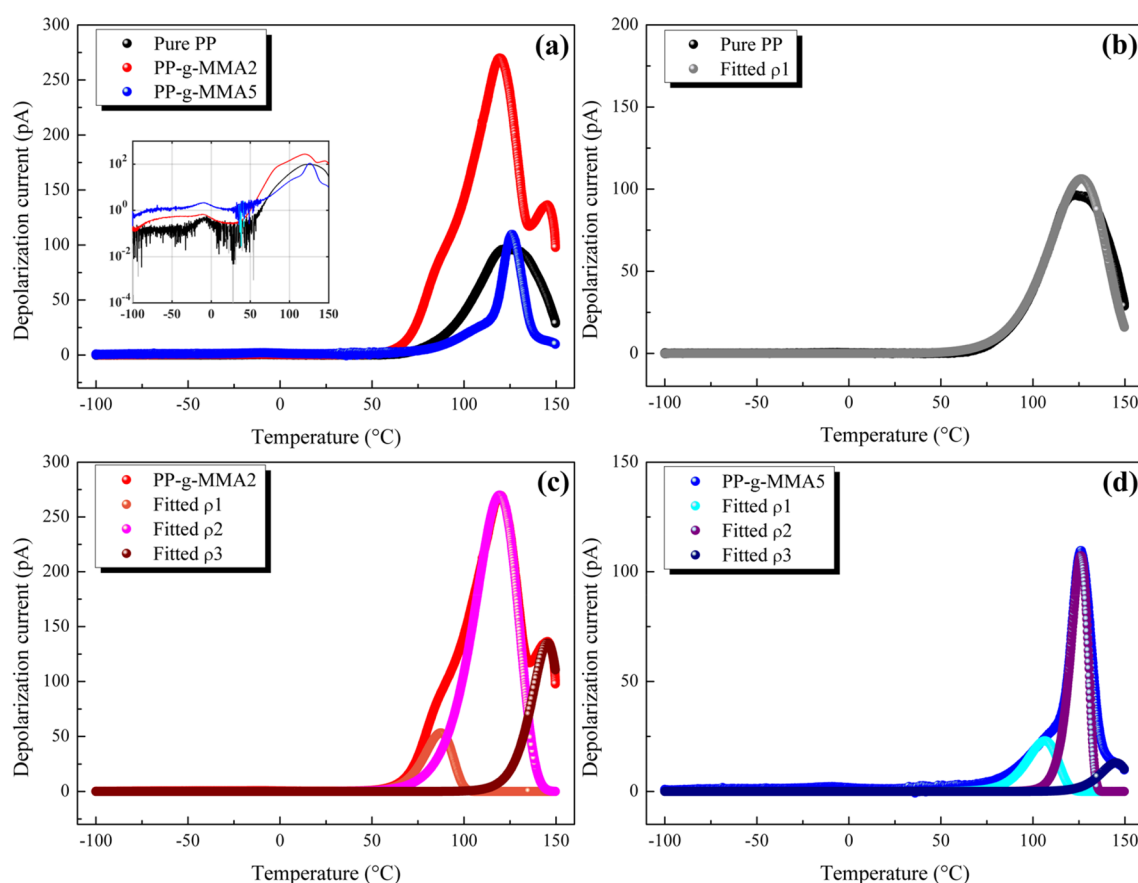


Figure 4. TSDC spectra of pure PP and MMA-grafted PP: (a) original spectra; (b)–(d) respective peak separation of the three TSDC spectra.

Table 1. Trap Parameters of Pure PP and MMA-Grafted PP

trap parameter	pure PP		PP-g-MMA2		PP-g-MMA5		
	peak ρ_1	peak ρ_1	peak ρ_2	peak ρ_3	peak ρ_1	peak ρ_2	peak ρ_3
trap level E_t (eV)	0.88	1.46	1.11	2.02	1.48	2.96	2.07
frequency factor ν (s^{-1})	3.57×10^8	1.44×10^{18}	7.71×10^{11}	1.26×10^{22}	2.79×10^{17}	2.31×10^{35}	5.54×10^{22}
peak temperature T_m ($^{\circ}C$)	126.35	87.36	119.33	145.40	106.46	126.08	145.11

peak can also be seen in the TSDC spectrum of poly methyl methacrylate (PMMA).²¹ These subzero peaks stem from the molecular segment relaxation. Because of the low polarity of PP matrix and low content of MMA polar side group, the depolarization current is also of a low level.

The current peaks under high temperature are believed to originate from charge detrapping. According to the single trap current equation (1), they are separated and quantitatively analyzed, denoted as ρ_1 – ρ_3 in the sequence of peak temperature. The peak separations and corresponding extracted parameters are exhibited in Figure 4b–d and Table 1. Pure PP, as shown in Figure 4b, only possesses one trap peak around 120 $^{\circ}C$, whose trap level is 0.88 eV, whereas for PP-g-MMA2 and PP-g-MMA5, three subpeaks can be observed. Peak ρ_2 of MMA-grafted PP appears in the same temperature range as peak ρ_1 of pure PP, whereas the trap level E_t and frequency factor ν are significantly higher; furthermore, they are proportional to the MMA content. Besides, two new peaks, i.e., ρ_1 and ρ_3 , emerge after grafting modification. Among them peak ρ_1 of the two grafted PPs shows close trap parameters except for the obvious difference in peak temperature T_m , whereas for peak ρ_3 , the peak location

and trap depth hardly show distinction between PP-g-MMA2 and PP-g-MMA5.

PP is a semicrystalline polymer. Many investigations indicate that the charge traps in PP originate from the amorphous areas and crystalline–amorphous boundaries, especially the defects or impurities.^{4–6,22} Physical defects such as cavities and pores would introduce shallow traps, whereas chemical defects such as broken chemical bonds and polar groups would generate deep traps.^{22–24} Therefore, only peak ρ_1 in pure PP can be ascribed to the traps generated by the molecular interactions in the disordered areas (i.e., amorphous areas and boundaries). Concerning MMA-grafted PP, the grafted groups/chains would act as impurities and thus would be extruded along the boundaries or into the amorphous phases during the crystallization process.¹⁵ The high polarity of the MMA group will strengthen the interaction with the PP molecular chain, thus deepening the trap level. Compared with PP-g-MMA2, the more polar MMA groups in PP-g-MMA5 raise the opportunity of interactions between the PP chain and the MMA group, interacting more sufficiently, thus presenting a higher trap level.

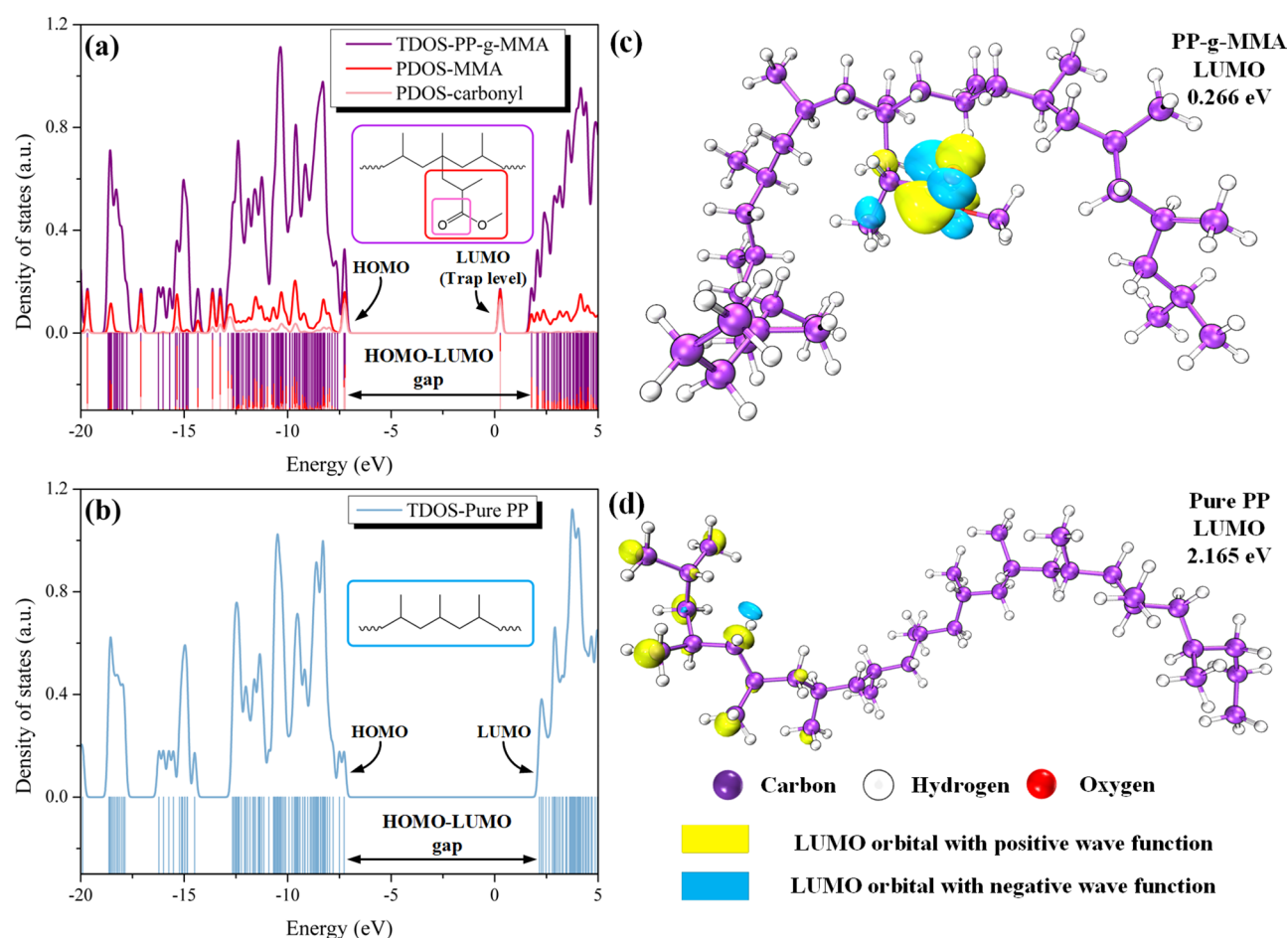


Figure 5. Density of electron states (DOS) spectra of (a) PP-g-MMA segment and (b) PP segment. The spectrum color corresponds to the frame color of chemical structure. LUMO orbitals are shown in (c) PP-g-MMA segment and (d) PP segment.

Peak ρ_1 of MMA-grafted PP, with a relatively low peak temperature T_m , seems to be an affiliated peak of the peak ρ_2 . This implies it would be connected to a more probable thermo-stimulated detrapping process, which may be an incidental process to the molecular interaction. The interaction among grafted MMA groups/chains might be a possible mechanism of this peak. The higher T_m in PP-g-MMA5 can be attributed to the tighter and stronger interaction because of the high MMA content.

Compared with other peaks, the trap parameters of peak ρ_3 are more stable, with the highest stimulation temperature. This phenomenon implies this peak may have nothing to do with the thermodynamics of molecules. The electronic relaxation of the polar chemical group, i.e., the carbonyl group, might be the origin. A further investigation with the assistance of computational analysis would be presented in the next chapter.

Besides the discussion above, there are also some noteworthy issues with respect to the TSDC spectra. For one thing, the peak current value does not show a clear trend among these peaks. This could be ascribed to the intrinsic randomness of TSDC characterization. The current value would float within a fairly large range even with respect to the same material but different samples, let alone different materials. A common understanding believes that the peak value of TSDC can reflect the trap density. However, in fact, the sufficiency of the trapping-detrapping process, as well as the dielectric performance (including AC permittivity and DC conductivity)

can also affect the TSDC and should be considered. Therefore, the peak current value is not a reliable index of the trap characteristics in our research.

For another thing, it can be seen in Table 1 that the peak temperature T_m and trap level E_t are not always proportional, which is contradictory to the common understanding on the relationship between T_m and E_t . This can be explained by eq 2 among the T_m , E_t , and frequency factor ν ; ¹⁴ i.e., the differential of eq 1 at T_m equals 0.

$$\frac{E_t}{kT_m^2} = \frac{\nu}{\beta} \exp\left(-\frac{E_t}{kT_m}\right) \quad (2)$$

Apparently, T_m and E_t are not of a simple positive correlation but are strongly connected with ν . Also, E_t and ν are affected by the peak shape. Generally speaking, E_t is proportional to ν and influenced simultaneously by T_m and the half-peak width (more directly: peak steepness). Either a higher T_m or a smaller half-peak width (steeper peak) can lead to a deeper trap level. The changes of the peak shape and corresponding parameters may imply the variation of the subject or process of detrapping/relaxation, which needs more in-depth investigations.

3.2. Computational Analysis. The computational analysis based on density functional theory (DFT) is processed upon the PP molecular segment, which can reflect the electron state distribution of the PP segment, thus revealing the chemical trap characteristics and probable electronic relaxation. Figure 5

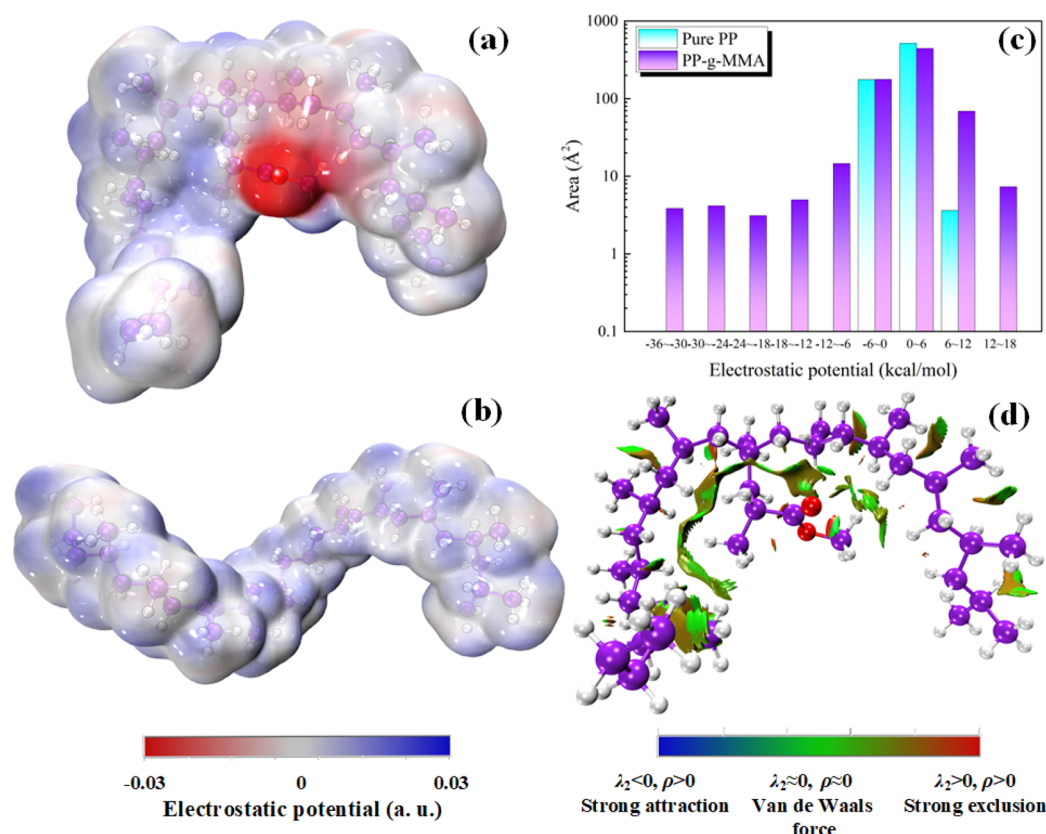


Figure 6. Surface electrostatic potential (ESP) of (a) PP-g-MMA segment and (b) PP segment. (c) ESP area distribution of the two segments. (d) Noncovalent interactions distribution based on reduced density gradient (RDG) theory.

panels a and b show the density of electron states (DOS) of PP and PP-g-MMA segments. To ascertain the contribution of different chemical structure (especially the grafted MMA group) to the electron states, the partial DOS (PDOS) spectra of MMA and carbonyl are also exhibited and compared with the total DOS (TDOS) of PP-g-MMA.

Compared with the pure PP segment, PP-g-MMA shows no obvious change with respect to the energy level of highest occupied molecular orbitals (HOMO). However, a new energy level, which corresponds to the LUMO of PP-g-MMA, emerges inside the gap. This can be regarded as the electron trap level of the PP-g-MMA. Figure 5 panels c and d depict the LUMO orbitals of PP and PP-g-MMA segments, whose energy levels are 2.165 and 0.266 eV, respectively. It can be seen that, after grafting MMA, the LUMO orbitals entirely “transfer” from the methyl and secondary carbon atom to the MMA group, particularly concentrated around the carbonyl. The PDOS of MMA and carbonyl in Figure 5a also verify that the new LUMO of PP-g-MMA is almost entirely contributed by the electron orbitals of the grafted group. Therefore, it can be confirmed that the grafted MMA group can introduce a deep trap level at the molecular level. Besides, the electron “transfer” energy, namely the trap depth can be calculated from the difference between the two LUMO energy values, i.e., 1.90 eV. Interestingly, it is very close to the trap level E_t of peak ρ_3 (2.02 and 2.07 eV) mentioned in TSDC analysis. Therefore, it is a strong support of the speculation that the origin of emerging peak ρ_3 in MMA-grafted PP is the MMA group, particularly the carbonyl structure.

Considering the high polarity of carbonyl in the grafted MMA group, it is easily supposed that the surface electrostatic

potential (ESP) of the PP molecule would be affected by grafting modification. Figure 6 panels a and b exhibit the Van der Waals surface ESP distribution of the two segments. A remarkable high negative potential area can be observed around the carbonyl. The calculated ESP are statistically analyzed and reflected in a histogram, as shown in Figure 6c. Because of the low polarity, the PP segment shows a very limited range of ESP from -6 to +12 kcal/mol. The maximum and minimum are +6.639 and -3.739 kcal/mol, respectively, whereas for PP-g-MMA, ESP range broadens, and high ESP areas (especially negative) are introduced by the grafted MMA group, with the maximum of +16.75 kcal/mol and minimum of -33.5 kcal/mol. The increment of high ESP areas implies more potential wells are introduced, thus hindering the positive/negative charge transportation, which would be beneficial to the macroscopic dielectric performance. Additionally, because of the grafted MMA, the Van der Waals surface area is also raised compared with that of unmodified PP (surface area increment about 111.433 bohr² in this case, with almost all of high ESP). Hence the molecular overlaps and interactions can also be enhanced, leading to more and deeper traps, as discussed in the TSDC analysis chapter.

Besides ESP, electron density can also be utilized to analyze the noncovalent interactions based on reduced density gradient (RDG) theory.²⁰ The RDG definition is presented as eq 3

$$\rho_{\text{RDG}}(r) = \frac{1}{2(3\pi^2)^{1/3}} \cdot \frac{|\nabla\rho(r)|}{(\rho(r))^{4/3}} \quad (3)$$

where $\rho(r)$ is the electron density. In the area where noncovalent interaction exists, both $|\nabla\rho(r)|$ and $(\rho(r))^{4/3}$

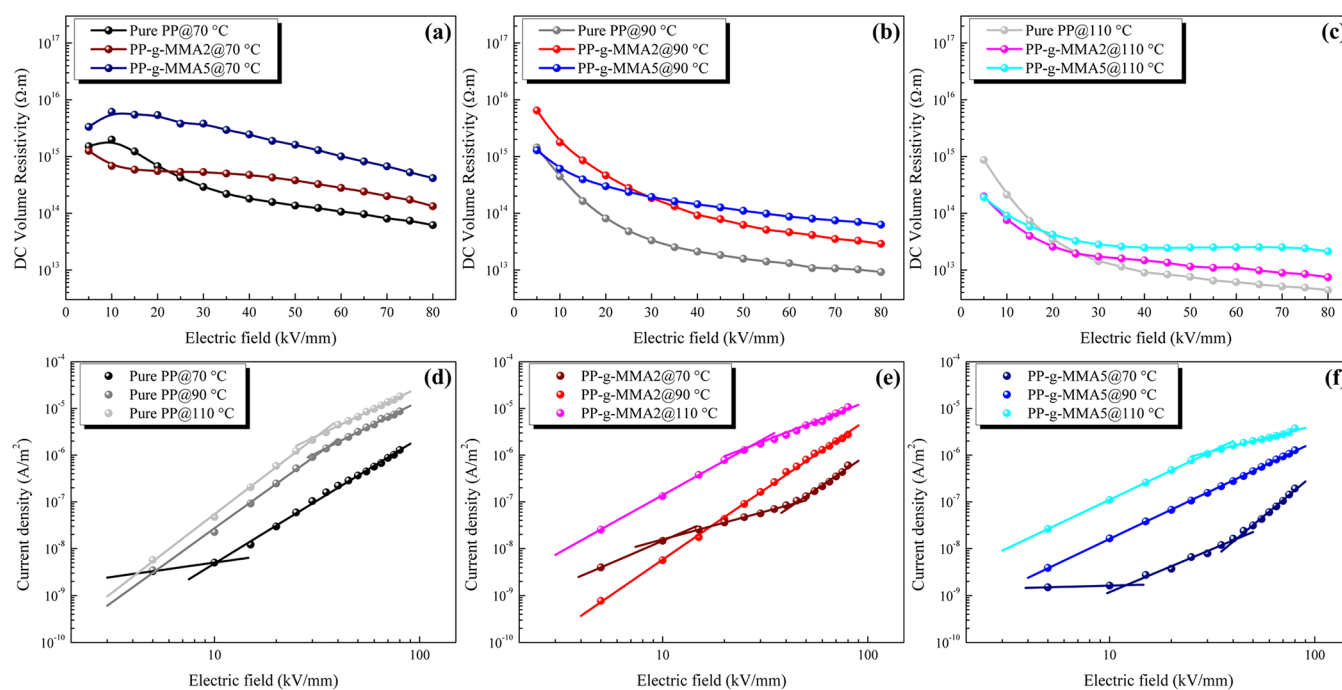


Figure 7. DC volume resistivity of pure PP and PP-g-MMA at (a) 70 °C, (b) 90 °C, and (c) 110 °C. Leakage current density versus electric field (J – E) characteristics of (d) pure PP, (e) PP-g-MMA2, and (f) PP-g-MMA5.

are very low, whereas ρ_{RDG} is of a moderate value. Generally, the ρ_{RDG} value in the noncovalent interaction area would not exceed 0.05.²⁰ When a contour surface is established with $\rho_{RDG} = 0.05$ throughout the space, the higher electron density ρ on the contour surface would indicate the stronger noncovalent interaction. Additionally, the type of interaction, attraction or exclusion, can be judged by the sign of the second largest eigenvalue λ_2 of the electron density Hessian matrix. As the color bar in Figure 6d describes, red color with a positive λ_2 refers to the exclusion, e.g., steric effect; blue color with the minus eigenvalue implicates the attraction, such as hydrogen bonds and halogen bonds; and green means very weak interactions where only Van der Waals force exists. It can be seen in Figure 6d that the contour surface envelops the grafted MMA. The brown color implies moderate exclusion exists between the PP chain and the MMA group, which may be ascribed to the steric effect. However, MMA is of aliphatic structure, whose steric occupation is not as significant as ring/aromatic structure. Hence the steric interaction is of a low level.

3.3. DC Volume Resistivity and SCLC Analysis. The DC volume resistivity under various electric field and high temperature are shown in Figure 7a–c. The resistivity enhancement of grafting modification is particularly significant under a high electric field. Moreover, the enhancement is proportional to the grafting content. This can be attributed to the reduced charge carrier mobility because of the charge trapping process in MMA-grafted PP. Considering the TSDC results, the macroscopic dielectric performances seem to be proportional to the trap depth, which had been reported in previous research on nanodielectrics.²⁵

Further investigations on the trap influence are carried out on the basis of space charge limited current (SCLC) theory.^{16,17} The J – E characteristics and linear fittings are shown in Figure 7d–f. In a nutshell, the J – E curve possesses two transition critical fields (denoted as E_1 and E_2) and three

corresponding periods/slopes k_1 to k_3 , which appear in turn along with rising electric field and temperature. However, these two transitions never appear simultaneously in one J – E curve under a certain temperature. $k_1 < k_2 > k_3$, meaning that the slope of J – E linear fitting first increases and then decreases. The first slope k_1 should correspond to the ohmic conduction, whereas k_2 and k_3 should be ascribed to two different SCLC conduction processes, which may be correlated with different trapping processes. Under 70 °C, The J – E curve shows the ohmic-SCLC transition, whose corresponding parameters are E_1 , k_1 , and k_2 . Grafting-modified PPs seem to possess one more period under low field (below 10 kV/mm). This is probably because of the ultralow current (approximately 0.5 pA in the case of PP-g-MMA5) affected by random noise and thus of low signal noise ratio (SNR). Therefore, it might be a “fake” datum and should be ignored. When it comes 90 °C, the J – E curve is “straightened”, with only one slope. It should be classified into SCLC conduction because its slope is close to the k_2 of 70 °C. Along with the further raised temperature to 110 °C, the curve comes to the transition with descending slope, i.e., k_2 to k_3 . It should be noticed that this transition has already occurred under 90 °C with respect to pure PP.

Table 2 exhibits all the linear fitting parameters of J – E characteristics. Under 70 °C, it can be seen that the transition critical electric field E_1 of MMA-grafted PP is significantly enhanced compared with that of pure PP, which indicates that the space charge injection in grafted PP is more difficult because of the barrier effect of deep traps near the electrode.^{4,7} However, regarding k_1 , PP-g-MMA5 > PP-g-MMA2 > pure PP does not strictly equals 1. This, on one hand, can be blamed on the random noise and low SNR of the ultralow leakage current under low field. On the other hand, some inevitable impurities and byproducts may be introduced in the grafting-modified PP and ionize during measurement, thus triggering ionic conduction. The ohmic conduction equation of single carrier can be expressed as (4)

Table 2. Slope and Critical Electric Field Parameters of the J – E Characteristics

designation	k_1	k_2	k_3	E_1 (kV/mm)	E_2 (kV/mm)
70 °C	pure PP	0.61	2.69	10.22	
	PP-g-MMA2	1.20	2.97	44.44	
	PP-g-MMA5	1.83	3.65	41.41	
90 °C	pure PP		3.16	2.22	34.94
	PP-g-MMA2		3.01		
	PP-g-MMA5		2.08		
110 °C	pure PP		3.35	2.09	31.82
	PP-g-MMA2		2.44	1.68	26.90
	PP-g-MMA5		2.09	1.10	31.42

$$J = qn\mu E \quad (4)$$

where q is the carrier charge quantity, n is the charge density, and μ is the charge mobility. Regarding ionic carriers, the density n might be proportional (maybe $n \propto E^\alpha$, $\alpha > 0$) to the electric field E . Hence k_1 deviates from 1 and increases with MMA content.

Regarding k_2 , the trend under 90 and 110 °C is pure PP > PP-g-MMA2 > PP-g-MMA5, whereas it is the opposite in the case of 70 °C. According to the SCLC theory and Child's law, the slope k of the trap-free (or uniformly distributed shallow traps) SCLC period should strictly equal 2,¹⁶ whereas in the case of nonuniform deep trap distribution, the slope would deviate from 2. In particular, assuming that deep trap distribution can be approximated by a characteristic temperature T_C , as expression 5 depicts¹⁶

$$n_t \propto \exp(-E_t/kT_C) \quad (5)$$

where n_t is the trap density, the J – E characteristics can be expressed as (6)¹⁷

$$J \propto e^{1-l} \left(\frac{\epsilon l}{H(l+1)} \right)^l \left(\frac{2l+1}{l+1} \right)^{l+1} \cdot E^{l+1} \quad (6)$$

where e is the electron charge (C), ϵ is the permittivity (F/m), H is the total trap density, and $l = T_C/T$. Here T is the temperature of measurement. T_C is a scale of the uniformity of trap distribution. Small T_C means the rapidly varying distribution, leading to low l thus low slope k in log–log diagram. Working backward on MMA-grafted PP, because of the chemical trap existence such as carbonyl trap, oxygen trap, the slope would significantly deviate from 2.²⁶ The introduced deep traps break the trap equilibrium, thus resulting in a lower T_C and slope. In addition, the average trap depth E_t in expression 5 is also raised, which is equivalent to lower T_C . That is to say, a deeper trap level corresponds to a lower slope k . This can explain the k_2 trends under 90 and 110 °C, which also coincide with the trend of the TSDC trap level E_t of peak ρ_2 . However, in the case of 70 °C, because of the high critical electric field, the charge injection in MMA-grafted PP is lower and not as sufficient as in pure PP. Although it possesses deeper traps, the trapped charge quantity is smaller and thus not dominant. In other words, the moderate temperature 70 °C cannot supply enough energy to completely stimulate the trapping–detrapping process in deep traps. Thus, the advantage of a deep trap is not obvious.

In the third period, the trapping injection is sufficient enough. Therefore, k_3 exhibits the same trend as the k_2 under 90 and 110 °C, which can be explained as well. k_3 is lower than

k_2 , i.e., a deeper trap level, implicating the deeper traps with higher activation energy (for example, peak ρ_3) sufficiently participate into the conduction process. It is noteworthy that in pure PP, the k_2 – k_3 transition has already occurred under 90 °C, which implies the charge injection process is more rapid than MMA-grafted PP and sooner reaches the transition state. Conversely speaking, the E_2 of MMA-grafted PP under 90 °C, if it exists, would exceed the highest electric field in our measurement. The higher k_3 in pure PP than those in grafted PPs indicates the relatively lower trap depth. These traps in pure PP may originate from the impurities and chemical defects, not as deep and stable as MMA-introduced traps.

4. CONCLUSIONS

This paper investigates the deep trap characteristic of MMA-grafted PP from three perspectives: The thermally stimulated depolarization current (TSDC) test indicates grafting modification can introduce deep traps. The original trap level of PP is deepened, which may be because of the molecular interactions between PP and grafted MMA groups. In addition, a new peak with high activation temperature emerges after grafting, whose energy level is approximately 2.0 eV. This is probably ascribed to the carbonyl structure.

Computational analysis based on quantum chemistry implicates that the grafted MMA group, particularly the carbonyl can introduce new electron orbitals inside the HOMO-LUMO gap of PP. The corresponding trap depth is 1.9 eV, very close to the result from TSDC. Moreover, the polar carbonyl can also introduce deep potential well thus affecting the charge transportation. The raised Van der Waals surface area would also enhance the molecular interactions.

The enhanced DC volume resistivity of MMA-grafted PP under high temperature is analyzed by space charge limited current (SCLC) theory. On one hand, MMA-grafted PP exhibits raised critical electric field of space charge injection. On the other hand, The SCLC period can be divided into two segments, probably corresponding to different trap characteristics. The decreased slope in MMA-grafted PP implies deeper trap level, which coincides with the trend exhibited in TSDC results. This study would offer some important insights into the trap characteristics of grafting-modified polypropylene.

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Notes

The authors declare no competing financial interest.

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REFERENCES

- (1) Hanley, T. L.; Burford, R. P.; Fleming, R. J.; Barber, K. W. A General Review of Polymeric Insulation for Use in HVDC Cables. *IEEE Electr. Insul. Mag.* **2003**, *19*, 13–24.
- (2) Wang, Y.; Zhou, X.; Chen, Q.; Chu, B. J.; Zhang, Q. M. Recent Development of High Energy Density Polymers for Dielectric Capacitors. *IEEE Trans Dielectr. Electr. Insul.* **2010**, *17*, 1036–1042.
- (3) Zhou, Y.; Hu, S. X.; Yuan, C.; Hu, J.; Li, Q.; He, J. L. Recyclable Polypropylene-based Insulation Materials for HVDC Cables: Progress and Perspective. *CSEE J. Power Energy Syst.* **2020**, DOI: 10.17775/CSEEJPES.2020.04740.
- (4) Yuan, H.; Hu, S. X.; Zhou, Y.; Yuan, C.; Song, W. B.; Shao, Q.; Shi, H. W.; Li, J.; Hu, J.; Li, Q.; et al. Enhanced Electrical Properties of Styrene-grafted Polypropylene Insulation for Bulk Power Transmission HVDC Cable. *CSEE J. Power Energy Syst.* **2021**, DOI: 10.17775/CSEEJPES.2021.00850.
- (5) Yuan, C.; Zhou, Y.; Zhu, Y. J.; Hu, S. X.; Liang, J. J.; Luo, Z.; Gao, B.; Zeng, T.; Zhang, Y. R.; Li, J.; et al. Improved High-temperature Electrical Properties of Polymeric Material by Grafting Modification. *ACS Sustain. Chem. Eng.* **2022**, *10*, 8685.
- (6) Zha, J. W.; Wu, Y. H.; Wang, S. J.; Wu, D. H.; Yan, H. D.; Dang, Z. M. Improvement of Space Charge Suppression of Polypropylene for Potential Application in HVDC Cables. *IEEE Trans Dielectr. Electr. Insul.* **2016**, *23*, 2337–2343.
- (7) Hu, S. X.; Dong, X. H.; Wang, W.; Huang, S. S.; Hu, J.; Li, Q.; He, J. L.; Shao, Q.; Yuan, H.; Shi, H. W.; et al. Dielectric Properties Improvement of Grafting-modified Polypropylene by Silane for HVDC Cable Insulation. *IEEE Trans Dielectr. Electr. Insul.* **2021**, *28*, 2004–2010.
- (8) Liu, H. L.; Du, B. X.; Xiao, M. Improved Energy Density and Charge Discharge Efficiency of Polypropylene Capacitor Film Based on Surface Grafting. *IEEE Trans Dielectr. Electr. Insul.* **2021**, *28*, 1539–1546.
- (9) Zhou, Y.; Hu, J.; Dang, B.; He, J. L. Mechanism of Highly Improved Electrical Properties in Polypropylene by Chemical Modification of Grafting Maleic Anhydride. *J. Phys. D: Appl. Phys.* **2016**, *49*, 415301.
- (10) Yuan, H.; Zhou, Y.; Zhu, Y. J.; Hu, S. X.; Yuan, C.; Song, W. B.; Shao, Q.; Zhang, Q.; Hu, J.; Li, Q.; et al. Origins and Effects of Deep Traps in Functional Group Grafted Polymeric Dielectric Materials. *J. Phys. D: Appl. Phys.* **2020**, *53*, 475301.
- (11) Zhou, Y.; Yang, J. M.; Zhao, H.; Sun, W. F.; Gao, M. Z.; Zhao, X. D.; Hu, M.; Xie, S. H. Improved DC Dielectric Performance of cPP-g-MAH/iPP/SEBS Composite with Chemical Graft Modification. *Materials* **2019**, *12*, 1094.
- (12) Hu, S. X.; Zhou, Y.; Yuan, C.; Wang, W.; Hu, J.; Li, Q.; He, J. L. Surface-modification Effect of MgO Nanoparticles on the Electrical Properties of Polypropylene Nanocomposite. *High Volt.* **2020**, *5*, 249–255.
- (13) Zhou, Y.; Yuan, C.; Li, C. Y.; Meng, P. F.; Hu, J.; Li, Q.; He, J. L. Temperature Dependent Electrical Properties of Thermoplastic Polypropylene Nanocomposites for HVDC Cable Insulation. *IEEE Trans Dielectr. Electr. Insul.* **2019**, *26*, 1596–1604.
- (14) Lei, Q. Q.; Wang, X.; Fan, Y. A New Method of Auto-separating Thermally Stimulated Current. *J. Appl. Phys.* **1992**, *72*, 4254–4257.
- (15) Hu, S. X.; Shao, Q.; Li, J.; Wang, M. T.; Huang, S. S.; Yuan, H.; Li, Q.; He, J. L. Quantitative Trap Analysis of Grafting-modified Polypropylene Based on Thermally Stimulated Depolarization Current. *Mater. Lett.* **2022**, *326*, 132990.
- (16) Rose, A. Space-charge-limited Current in Solids. *Phys. Rev.* **1955**, *97*, 1538–1544.
- (17) Mark, P.; Helfrich, W. Space-charge-limited Currents in Organic Crystals. *J. Appl. Phys.* **1962**, *33*, 205–215.
- (18) Lu, T.; Chen, F. W. Multiwfn: A Multifunctional Wavefunction Analyzer. *J. Comput. Chem.* **2012**, *33*, 580–592.
- (19) Lu, T.; Chen, F. W. Quantitative Analysis of Molecular Surface Based on Improved Marching Tetrahedra Algorithm. *J. Mol. Graph. Model.* **2012**, *38*, 314–323.
- (20) Johnson, E. R.; Keinan, S.; Mori-Sánchez, P.; Contreras-Garcia, J.; Cohen, A. J.; Yang, W. T. Revealing Noncovalent Interactions. *J. Am. Chem. Soc.* **2010**, *132* (18), 6498–6506.
- (21) Kalogerias, I. M.; Pallikari, F.; Vassilikou-Dova, A. The Diverse Effect of Antiplasticizer in the Molecular Dynamics of an Organic Dye-doped Polymer Observed at Different Motional Lengthscales. *Eur. Polym. J.* **2009**, *45*, 1377–1384.
- (22) Tian, F. Q.; Lei, Q. Q.; Wang, X.; Wang, Y. Effect of Deep Trapping States on Space Charge Suppression in Polyethylene/ZnO Nanocomposite. *Appl. Phys. Lett.* **2011**, *99*, 142903.
- (23) Peng, S. M.; Dang, B.; Zhou, Y.; Hu, J.; He, J. L. Functionalized TiO₂ Nanoparticles Tune the Aggregation Structure and Trapping Property of Polyethylene Nanocomposites. *J. Phys. Chem. C* **2016**, *120*, 24754–24761.
- (24) Zhai, J. T.; Li, W. K.; Zha, J. W.; Cheng, Q.; Bian, X. M.; Dang, Z. M. Space Charge Suppression of Polyethylene Induced by Blending with Ethylene-butyl Acrylate Copolymer. *CSEE J. Power Energy Syst.* **2020**, *6* (1), 152–159.
- (25) Li, S. T.; Min, D. M.; Wang, W. W.; Chen, G. Linking Traps to Dielectric Breakdown Through Charge Dynamics for Polymer Nanocomposites. *IEEE Trans Dielectr. Electr. Insul.* **2016**, *23*, 2777–2785.
- (26) Lei, Q. Q.; Liu, G. Y. How to Understand the Two Basic Physical Processes of Polarization and Conductance in Engineering Dielectrics and Scientific Principles and Methods of Their Measurement. *Proc. CSEE* **2018**, *38*, 6769–6789 (in Chinese).